

Naree Foundation & NF Venture Studio (NFVS)

Unified CRM, HRMS & Enterprise Operations Platform

A complete end-to-end overview of platform architecture, technology rationale, multi-tenant isolation, granular access control, functional modules, data models, and deployment protocols.

Target Organizations: Naree Foundation & NF Venture Studio	System Version: 1.0.0 (Production Release)
Document Classification: Confidential / Managerial Briefing	Audit Date: October 2026
Primary Framework: Next.js 16 (App Router) + React 19	Database: PostgreSQL (Supabase AWS) via Prisma ORM

1. Executive Summary & Business Objectives

1.1 Platform Purpose

The **NFVS CRM & HRMS Platform** is a centralized, cloud-native Enterprise Resource Planning (ERP) and Human Resource Management System engineered specifically to eliminate fragmented third-party SaaS subscriptions. It establishes a unified, secure portal powering day-to-day operations for both **Naree Foundation** (a non-profit social development organization) and **NF Venture Studio** (a commercial venture incubator).

1.2 Strategic Business Value & ROI

- **Zero Cross-Tenant Leakage:** Hosts two distinct corporate entities on a single unified cloud deployment while enforcing airtight organizational boundaries.
- **Absolute Peer Privacy & Data Isolation:** Standard employees cannot inspect peer tasks, managerial workflows, compensation structures, or internal operational agendas.
- **Strict Direct Line Manager Governance:** Implements a real-world managerial tree where Line Managers only delegate to and monitor their directly assigned subordinates.
- **Immutable Mutation Audit Trail:** Every record mutation (create, edit, delete, status change) is permanently logged with actor credentials, timestamps, and field diffs for governance compliance.

2. Complete Technology Stack & Architectural Rationale

Layer / Role	Technology	Version	Technical Purpose & Strategic Rationale
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Full-Stack Framework	Next.js (App Router)	16.3.4	Delivers React Server Components (RSC) for zero-client bundle data fetching, fast Server-Side Rendering (SSR), and unified API route handlers.
Frontend Engine	React	19.2.8	Modern concurrent rendering, fine-grained DOM hydration, and modular UI component lifecycles for high-performance dashboard widgets.
Language Foundation	TypeScript	5.x	End-to-end static type safety bridging database models, API payload contracts, and UI components to eliminate runtime type errors.
Design System	Tailwind CSS	4.x	Utility-first styling powering a bespoke dark-mode enterprise theme (Deep Slate #0a0f1d, Navy #0f172a, Emerald, Amber, Blue).
ORM & Modeling	Prisma ORM	6.19.3	Type-safe database abstraction managing 3,400+ lines of declarative schema models with automated migration synchronization.
Persistence Layer	PostgreSQL (AWS)	15+	ACID-compliant relational engine hosted on Supabase AWS. Utilizes PgBouncer on port 6543 for API traffic and port 5432 for schema migrations.
Input Validation	Zod	4.5.4	Strict runtime schema parsing and sanitization protecting all API boundaries from malformed or unauthorized field injections.
Security & Hashing	bcryptjs	3.0.3	Multi-round salted cryptographic password hashing safeguarding system credentials and employee account authentications.
Iconography	Lucide React	1.43.0	Crisp, tree-shakable SVG iconography optimized for navigation, system badges, and operational status chips.

3. System Architecture & Design Patterns

3.1 Layered Modular Monolith Architecture

The system employs a **Layered Modular Monolith** design pattern. This pattern delivers the rapid development and transactional simplicity of a single cohesive codebase while preserving clean domain separation between modules (HR, Tasks, Attendance, Payroll, and CRM):

1. PRESENTATION LAYER: Next.js Server Components (RSC) + Interactive Client Components + Tailwind CSS

↓ *Authenticated JSON Requests / HTTP-Only Cookies*

2. CONTROLLER & ROUTING LAYER: Next.js App Router API Handlers (/api/*) + Zod Schema Validation

↓ *Validated Business Invocations*

3. BUSINESS SERVICE LAYER: TaskService | EmployeeService | RbacService | AuditLogger | PayrollEngine

↓ *Type-Safe Relational Queries*

4. DATA ACCESS / ORM LAYER: Prisma ORM Client + PgBouncer Transaction Pooler (Port 6543)

↓ *ACID Relational SQL*

5. PERSISTENCE STORAGE LAYER: PostgreSQL Database Instance (Supabase Cloud Infrastructure)

3.2 Architectural Highlights & Reliability Features

- **Stateless Route Handlers:** Endpoints authenticate per request using secure JWT/cookies and dynamic headers, enabling horizontal scaling without sticky sessions.
- **Connection Pooling:** Utilizes PgBouncer transaction-mode connection pooling to handle high concurrency without overwhelming PostgreSQL process limits.

- **Deterministic Hydration Protection:** Date and time values utilize explicit server-matched locale formatting (`en-US` / `en-IN`) and hydration guards to eliminate SSR client-render mismatches.

4. Multi-Tenancy & Corporate Boundary Isolation

A central requirement of the system is the simultaneous operation of **Naree Foundation** and **NF Venture Studio** within a unified platform without compromising data confidentiality or legal entity independence.

- **Shared Database, Row-Level Logical Isolation:** All operational records (Employees, Tasks, Leaves, Clients, Attendance) include a mandatory foreign key organizationId referencing the Organization master table.
- **Context Resolution Middleware:** API requests automatically inspect the active tenant via the x-company-id header or user session. Queries strictly filter by where: { organizationId } to prevent cross-tenant record leakage.
- **Real-Time Tenant Switcher:** Authorized executive roles (Super Admin, Board, CEO) can seamlessly toggle their operational workspace between Naree Foundation and NF Venture Studio via the global header switcher.

5. Role-Based Access Control (RBAC) & Security Governance

The platform enforces a granular, numeric hierarchy with 9 distinct operational privilege tiers:

Level	Role Code	Role Title	Operational Permissions & Data Visibility Scope
100	SUPER_ADMIN	Global Super Administrator	Complete global platform oversight, role configuration, tenant creation, and system audit logs. Operates with non-interfering read-only operational visibility.
90	ADMIN	Platform Administrator	Full organization-wide management, CXO account oversight, and global system configurations.
80	CHAIRPERSON	Board Chairperson	Strategic executive visibility, board-level performance analytics, cross-entity compliance audits.
75	CEO	Chief Executive Officer	Full organizational operational authority. Dispatches directives to CXOs, approves high-level milestones.
70	C-SUITE / HR	COO, CFO, CIO, CTO, CMO, HR	Departmental leadership. HR manages full employee lifecycles, dossiers, compensation, and onboarding dispatches.
50	DEPARTMENT_HEAD	Department Head	Supervises departmental workflows, approves leaves, and oversees cross-team deliverable timelines.
30	MANAGER	Direct Line Manager	Direct subordinate task tracking, work delegation, and status oversight for assigned direct reports only.
10	EMPLOYEE	Staff Member	Strict self-access: Personal assigned tasks, personal profile, own attendance records, and personal payslips. Absolutely no peer visibility.

6. Hierarchical Task Delegation & Direct Line Manager System

6.1 The Operational Problem Solved

In standard corporate CRM/task boards, all team members frequently see each other's tasks, leading to privacy breaches, inter-departmental gossip, and exposure of sensitive operational assignments. The NFVS CRM implements a strict **Attribute-Based + Managerial Hierarchy Access Control (ABAC)** rule set.

6.2 Enforced Security & Delegation Rules

- Standard Staff Peer Isolation:** Regular employees (Level < 30) can ONLY query and inspect tasks where they are the explicit Assignee (assigneeId == employeeId) or Creator. They have zero visibility into peer tasks or manager workflows.
- Direct Line Manager Scoping:** A Line Manager (Level 30) can view: (a) Their own tasks, (b) Tasks they authored, and (c) Tasks assigned to subordinates who explicitly report to them (assignee.managerId == managerId). Line Managers cannot view tasks of employees reporting to other managers.
- Restricted Delegation Controls:** When a Line Manager creates or reassigns a task, the assignee dropdown dynamically filters to display *only their direct line reports*, preventing unauthorized workload dumping onto other teams.
- Executive Global Visibility:** High-level leadership (CEO, HR, Admin, Super Admin) maintains global cross-team audit visibility for comprehensive oversight.

7. Core Functional Modules & Feature Breakdown

Module	Core Capabilities & Technical Specifications
7.1 Employee Lifecycle & 10-Section Dossier	Comprehensive 10-tab employee master: (1) Personal Demographics, (2) Official Employment (ID, Designation, Department, Line Manager, Work Mode), (3) Permanent & Current Address, (4) Government IDs (Aadhaar, PAN, Passport), (5) Bank & Direct Deposit Details, (6) Emergency Contacts, (7) Education History, (8) Experience & Skills, (9) Document Soft Copies, and (10) Corporate Credentials & Role.
7.2 Document Management Vault (10MB)	Encrypted soft-copy document repository supporting PDF, PNG, JPG, and DOCX files up to 10MB per document. Maintains verification lifecycle states: PENDING_VERIFICATION, VERIFIED, and REJECTED with audit notes.
7.3 Task Management Engine	Multi-view operational board featuring Kanban lanes (TODO, IN_PROGRESS, BLOCKED, COMPLETED, CANCELLED), priority tagging (LOW, MEDIUM, HIGH, URGENT), client/project linking, and real-time nested discussion threads.
7.4 Attendance & Shift Tracking	Daily check-in / check-out timestamps with client IP address logging, break tracking, total work duration computation, and automated late/half-day flagging.
7.5 Leave & Absence Governance	Multi-category leave balances (Casual, Sick, Paid, Maternity). Multi-step approval routing through direct line manager and HR, integrated with organizational holiday calendars.
7.6 Payroll & Compensation Automation	Granular salary structures (Basic Pay, HRA, Special Allowance, PF, PT, Professional Tax). Monthly payroll period generation, tax deductions, and automated printable payslip generation.
7.7 Client CRM & Pipeline	Corporate client registry, opportunity deal pipeline tracking (LEAD, PROSPECT, PROPOSAL, NEGOTIATION, WON, LOST), interaction logs, and meeting schedule widgets.
7.8 Operations & Duty Dispatch	Field operations dispatching (OnDutyAssignment) with real-time status (SCHEDULED, ACTIVE, COMPLETED, INCIDENT_FLAGGED) and automated employee availability monitoring.
7.9 Mutation Audit Logging	Tamper-evident audit engine capturing actor identity, target entity, action type (CREATE, UPDATE, DELETE), IP address, timestamp, and field-level change diffs.

8. Database Schema & Data Model Specifications

The relational schema is maintained through Prisma ORM across 3,499 declarative lines. Below is the core relational topology:

```
[Organization: Tenant Master (Naree / NFVS)]
|-- 1:N --> [Department] (HR, Operations, Finance, Tech, Incubation)
|-- 1:N --> [Employee] <-----+ (Self-referential managerId hierarchy)
|
|   |-- 1:1 --> [User] (Authentication credentials & Role level)
|   |-- 1:1 --> [EmployeeProfile] (10-Section Personal Dossier)
|   |-- 1:N --> [EmployeeDocument] (Soft copies up to 10MB)
|   |-- 1:N --> [AttendanceRecord] (Clock events, work hours)
|   |-- 1:N --> [LeaveRequest] (Approval workflow)
|   |-- 1:N --> [SalaryStructure] & [PayrollRecord]
|   +-- 1:N --> [Task] (Assignee / Creator)
|-- 1:N --> [Client] -- 1:N --> [Opportunity] (CRM Deals & Pipelines)
|-- 1:N --> [Operation] (Field missions & on-duty dispatches)
+-- 1:N --> [AuditLog] (Immutable compliance mutation history)
```

9. API Architecture & Endpoint Catalog

HTTP Route	Methods	Auth & Scoping	Functionality & Business Rules
/api/auth/login	POST	Public	Validates corporate credentials against bcrypt hash; issues secure HTTP-only session cookies.
/api/auth/me	GET	Authenticated	Returns active user identity, numeric role level, employee link, and active company context.
/api/employees	GET, POST	HR / Admin / CXO	Lists paginated employees with department filters; registers new employees with direct line manager assignment.
/api/employees/[id]	GET, PATCH, DELETE	RBAC Scoped	Retrieves full dossier; updates master fields; performs administrative deactivation/deletion.
/api/employees/[id]/profile	GET, PUT	Self / HR	Retrieves and saves 10-section profile information; recalculates profile completion score.
/api/employees/[id]/documents	GET, POST	Self / HR	Uploads soft-copy identity and qualification documents up to 10MB; updates document verification state.
/api/tasks	GET, POST	Authenticated (Scoped)	Fetches tasks strictly isolated by Line Manager and Employee hierarchy; dispatches new tasks.
/api/tasks/[id]	GET, PATCH, DELETE	Task Authorized	Updates task status (TODO → COMPLETED), modifies priority, updates deadlines.
/api/attendance	GET, POST	Authenticated	Logs real-time employee check-in / check-out timestamps, IP addresses, and retrieves monthly attendance logs.
/api/payroll/generate	POST	HR / CFO / Admin	Executes automated batch payroll generation and populates individual employee monthly payslips.

10. Deployment, Infrastructure & Operational Runbook

10.1 Environment Variables Architecture

- DATABASE_URL: High-throughput transaction-mode connection pooler via PgBouncer on port **6543** for all runtime API traffic.
- DIRECT_URL: Direct session connection to PostgreSQL on port **5432** for executing DDL schema migrations with Prisma CLI.

10.2 Production Release Sequence

```
# 1. Install locked dependencies
npm install

# 2. Re-generate Prisma Client TypeScript bindings
npx prisma generate

# 3. Execute database schema migrations
npx prisma migrate deploy

# 4. Build optimized Next.js production bundle
npm run build

# 5. Start production runtime server
npm run start
```

Prepared By: Engineering & Operations Team	Approved By: Executive Leadership (Naree & NFVS)
Target Audience: Internal Management & Technical Leads	Release State: Verified Production Ready